MOHAMMED LAKHAL



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FULL STACK JAVA ENGINEER | SPRING BOOT | REACT | MICROSERVICES

FULL STACK JAVA ENGINEER

Software engineer specializing in Full Stack development, with experience in designing banking applications and SaaS platforms. Expertise in Java, Spring Boot, microservices architectures, and secure REST APIs. Proficient in React.js, PostgreSQL, MongoDB, Spring Batch, and Keycloak (OAuth2/JWT), as well as Docker and CI/CD pipelines. Committed to code quality, security, and application reliability.

EDUCATION

Marrakech Private University

ICTSE Engineer – IT, Information Systems — 2022–2024

Engineering education focused on software development, modern architectures, information systems, and the design of high-performance, secure, and scalable IT solutions.

Marrakech Private University

Bachelor's Degree in Computer Science — 2019–2021

Training covering the fundamentals of computer science, including programming, databases, networks, operating systems, and software engineering.

EXPERIENCE

Projects: Zatsit — May 2026 – Present

Full Stack Java / React Developer — Freelance (Remote)

Project 1 — Digital Engagement | Banking Platform for Segmentation and Campaigns

Design and development of a banking "Digital Engagement" platform dedicated to customer segmentation, defining targeted audiences, and managing personalized campaigns.
Full-stack development using Java 17, Spring Boot, and React.js within a microservices architecture based on Spring Cloud and OpenFeign.
Design of microservices for segment definition, membership management, and the definition, calculation, and materialization of audiences.
Implementation of asynchronous and batch processing—including pagination, scheduling, and error-recovery mechanisms—for audience calculation.
Development of secure REST APIs using Spring Security, OAuth2/JWT, and Keycloak, documented via OpenAPI/Swagger.
Data management using PostgreSQL and MongoDB, with inter-service communication implemented via OpenFeign.
Development of automated tests using JUnit 5, Mockito, and Testcontainers; integration of observability tools including Micrometer, Prometheus, OpenTelemetry, and Spring Actuator.
Contribution to containerization and deployment automation using Docker, GitLab CI/CD, and Jenkins.

Project 2 — Digital Assist | Banking Solution

Development of a "Digital Assist" banking platform using React.js, TypeScript, and Spring Boot.
Design and development of microservices and secure REST APIs to support platform functionality.
Implementation of features for customer segmentation and the creation of targeted audiences.
Development of capabilities for managing and personalizing multi-channel campaigns.
Integration of Keycloak for authentication, role management, and access control.

March 2025 – May 2026
Marrakech

Full Stack Developer | KAZ-IA

Project 1 — Multi-tenant SaaS CRM Platform

Development of a SaaS CRM platform using Java, Spring Boot, and React.js, based on a microservices architecture.
Design and development of secure REST APIs for the platform's business functionalities.
Integration of Keycloak for identity management, authentication, and role-based access control (RBAC).
Data modeling using PostgreSQL, with the implementation of data isolation mechanisms between tenants.
Containerization of services using Docker and automation of CI/CD pipelines via GitLab CI/CD.

Project 2 — Intelligent Document Extraction Platform

Development of a web application using Next.js, React.js, and Node.js for processing and extracting data from documents.
Integration of an artificial intelligence engine to extract and structure data from documents.
Design of dynamic document templates and generation of structured exports.
Management and persistence of document data using MongoDB.
Enhancement of the application's modularity and maintainability through a clear structuring of components and features.

February 2024 – August 2024 Casablanca

Final-Year Internship – Full Stack Developer | CIH BANK

- Design and development of a back-office application enabling Level 2 (N2) support teams to view and track IBM MQ messages, featuring a Spring Boot backend and a React.js interface.
- Development of a batch process using Spring Batch to automate the bulk creation of third-party accounts.
- Development of REST services for the geolocation of bank branches and the automation of internal processes.

Technologies: Java, Spring Boot, Spring Security, Spring Data JPA, Spring Batch, IBM MQ, REST API, React.js, JavaScript, Material UI, Keycloak, PostgreSQL.

SKILLS

Backend :
- Java
- Spring Boot
- Spring Security
- Spring Data
- JPA Hibernate,
- Spring Batch
- Microservices
- API REST

Frontend :
- React.js
- Next.js
- TypeScript,
- JavaScript

Databases:
- PostgreSQL
- MongoDB
- MySQL,
- SQL

DevOps & CI/CD:
- Docker
- GitLab CI/CD
- Jenkins
- GitHub Actions
- Git

Messaging:
- IBM MQ

Security & IAM
- Keycloak
- OAuth2
- JWT

Testing & Observability
- Prometheus
- Micrometer
- OpenAPI/Swagge
- Testcontainers
- Mockito
- JUnit 5
- OpenTelemetry

Tools & Quality:
- Maven
- SonarQube,
- GitHub, GitLab

Methodologies:
- Agile, Scrum

Concepts
- Programmation orientée objet (POO)